

MATH 281 Applied Boundary Value Problems

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1 Heat Equation

Definition 1.1. A **partial differential equation** (PDE) is an equation containing partial derivatives.

1.1 Conduction of Heat in a 1-D Rod

If we have 1 dimensional rod with cross sectional area A , L , and thermal energy density $e(x, t)$, the heat energy of a slice on the rod from x to $x + \Delta x$ is:

$$\text{heat energy} = e(x, t)A\Delta x$$

If there is no lateral heat flow on the rod, then the rate of change of the heat energy of this slice must be the sum of the net heat flux into the slice and the heat generated in the slice:

$$\frac{\partial}{\partial t}[e(x, t)A\Delta x] \approx \phi(x, t)A - \phi(x + \Delta x, t)A + Q(x, t)A\Delta x$$

Where $Q(x, t)$ is the heat energy generated per unit volume.

We can easily simplify this by dividing by $A\Delta x$ and taking the $\lim_{\Delta x \rightarrow 0}$ to get the following PDE:

$$\frac{\partial e}{\partial t} = -\frac{\partial \phi}{\partial x} + Q$$

We can also derive this equation by considering a finite slice from $x = a$ to $x = b$. The rate of change of heat energy of this slice is the **integral conservation law**:

$$\frac{d}{dt} \int_a^b e dx = \phi(a, t) - \phi(b, t) + \int_a^b Q dx$$

We can convert this to a single integral by noting that $\frac{d}{dt} \int_a^b e(x, t) = \int_a^b \frac{\partial e(x, t)}{\partial t}$ and that $\phi(a, t) - \phi(b, t) = -\int_a^b \frac{\partial \phi}{\partial x} dx$, resulting in:

$$\int_a^b \left(\frac{\partial e}{\partial t} + \frac{\partial \phi}{\partial x} - Q \right) dx = 0$$

Since this must be true for any arbitrary a and b , the integrand must be zero:

$$\frac{\partial e}{\partial t} = -\frac{\partial \phi}{\partial x} + Q$$

Because we also describe the temperatures of materials instead of heat energy, it is useful to convert our equations. We let:

$$\begin{aligned} u(x, t) &\equiv \text{temperature} \\ c &\equiv \text{specific heat capacity} \\ \rho &\equiv \text{density} \end{aligned}$$

Then, the thermal energy density can be expressed as:

$$e(x, t) = c(x)\rho(x)u(x, t)$$

Therefore, the conservation of thermal energy can be expressed as:

$$c(x)\rho(x)\frac{\partial u}{\partial t} = -\frac{\partial \phi}{\partial x} + Q$$

Law 1.1. Fourier's law of heat conduction

$$\phi = -K_0 \frac{\partial u}{\partial x}$$

where K_0 is the thermal conductivity of the material.

Substituting this in, we get:

$$c\rho \frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(K_0 \frac{\partial u}{\partial x} \right) + Q$$

If c , ρ , and K_0 are all constant, then this simplifies to the **heat equation**:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

where

$$k = \frac{K_0}{c\rho}$$

is the **thermal diffusivity**.

We can similarly derive integral and regular diffusion laws for diffusion of a chemical:

$$\begin{aligned} \frac{d}{dt} \int_a^b u(x, t) dx &= \phi(a, t) - \phi(b, t) \\ \frac{\partial u}{\partial t} + \frac{\partial \phi}{\partial x} &= 0 \end{aligned}$$

where $u(x, t)$ is the density or concentration of the chemical per unit volume.

Law 1.2. Fick's Law of Diffusion

$$\phi = -k \frac{\partial u}{\partial x}$$

where k is chemical diffusivity.

Using this, we get the **diffusion equation**:

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2}$$

1.2 Initial and Boundary Conditions

Since the heat equation has one time derivative, we need one **initial condition** (IC). Additionally, we need one **boundary condition** (BC) at each end of the rod.

One potential boundary condition is having a set temperature, called the **prescribed temperature**, at a boundary:

$$u(0, t) = u_B(t)$$

Definition 1.2. A **Dirichlet condition** is one where the values of a function at a boundary are specified.

We can also specify heat flow at a boundary:

$$-K_0(0) \frac{\partial u}{\partial x}(0, t) = \phi(t)$$

The simplest case of this is where the boundary is insulated:

$$\frac{\partial u}{\partial x} = 0$$

Definition 1.3. A **Neumann condition** is one where the derivative $\frac{\partial u}{\partial x}$ at a boundary is specified.

Finally, we can allow the boundary to exchange heat with the surrounding.

Theorem 1.1. Newton's Law of Cooling The rate of heat transfer between a body and its surroundings can be approximated by the following:

$$-K_0(0) \frac{\partial u}{\partial x}(0, t) = -H[u(0, t) - u_B(t)]$$

where H is called the **heat transfer coefficient**.

1.3 Equilibrium Temperature Distribution

1.3.1 Prescribed Temperatures

Definition 1.4. An **equilibrium** or **steady-state** solution to a partial differential equation is one that does not depend on time.

Suppose we have a simplified scenario where the thermal conditions are constant, no heat is generated, and heat is prescribed on the boundaries:

$$\begin{aligned} u(0) &= T_1 \\ u(L) &= T_2 \end{aligned}$$

We can easily find the equilibrium solution by setting the rate of heat flow to zero, double integrating, and solving:

$$\begin{aligned} \frac{d^2 u}{dx^2} &= 0 \\ \rightarrow u(x) &= C_1 x + C_2 \end{aligned}$$

We then use the boundary conditions to solve for C_1 and C_2 :

$$\begin{aligned} u(0) &= C_2 = T_1 \\ u(L) &= C_1 L + T_1 \rightarrow C_1 = \frac{T_2 - T_1}{L} \end{aligned}$$

Thus, we have the following solution:

$$u(x) = T_1 + \frac{T_2 - T_1}{L}x$$

1.3.2 Insulated Boundaries

If instead the boundaries are insulated, we have the following PDE:

$$\begin{aligned} \text{PDE: } \frac{\partial u}{\partial t} &= k \frac{\partial^2 u}{\partial x^2} \\ \text{IC: } u(x, 0) &= f(x) \\ \text{BC1: } \frac{\partial u}{\partial x}(0, t) &= 0 \\ \text{BC2: } \frac{\partial u}{\partial x}(L, t) &= 0 \end{aligned}$$

We similarly solve for the equilibrium solution:

$$\begin{aligned} \text{ODE: } \frac{d^2 u}{dx^2} &= 0 \\ \text{BC1: } \frac{du}{dx}(0) &= 0 \\ \text{BC2: } \frac{du}{dx}(L) &= 0 \\ \rightarrow u &= C_1 x + C_2 \rightarrow C_1 = 0 \rightarrow u = C_2 \end{aligned}$$

Since the boundaries are insulated, the total energy in the rod must be constant, so:

$$\begin{aligned} \int_0^L c \rho u dx &= \int_0^L c \rho f(x) dx \\ \rightarrow u(x) &= C_2 = \frac{1}{L} \int_0^L f(x) dx \end{aligned}$$

2 Method of Separation of Variables

2.1 Linearity

Definition 2.1. An operator L is **linear** (and thus called a **linear operator**) if:

$$L(c_1 u_1 + c_2 u_2) = c_1 L(u_1) + c_2 L(u_2)$$

for any functions u_1 and u_2 and constants c_1 and c_2 .

Definition 2.2. A **linear equation** for the unknown u is of the form:

$$L(u) = f$$

where L is a linear operator and f is known.

Definition 2.3. A linear equation $L(u) = f$ is called **homogeneous** if $f = 0$. Otherwise, it is called **nonhomogeneous**.

Theorem 2.1. Principle of Superposition If u_1 and u_2 satisfy an arbitrary linear homogenous equation, then any arbitrary combination $c_1 u_1 + c_2 u_2$, where c_1 and c_2 are constants, also satisfies the same equation.

2.2 Heat Equation with Zero Temperature at Finite Ends

Suppose we have a rod with fixed temperatures of 0 at the boundaries, represented by the following:

$$\begin{aligned} \text{PDE: } & \frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} \quad 0 < x < L, \quad t > 0 \\ \text{BC: } & u(0, t) = 0 \\ & u(L, t) = 0 \\ \text{IC: } & u(x, 0) = f(x) \end{aligned}$$

We can solve this using the **method of separation of variables**.

2.2.1 Product Solutions

The first step is to determine a **product solution** of the form:

$$u(x, t) = H(x)G(t)$$

Substituting this into the PDE and dividing by $kH(x)G(t)$ to separate variables gives us:

$$\frac{1}{kG} \frac{dG}{dt} = \frac{1}{H} \frac{d^2 H}{dx^2}$$

For this to be true for any x and t , both sides must equal a constant, which we set as $-\lambda$, where λ is the **separation constant**. This gives us two ordinary differential equations, one for $H(x)$ and one for $G(t)$.

The product solutions must satisfy the boundary conditions. $u(0, t) = 0$ implies that either $H(0) = 0$ or $G(t) \equiv 0$. The latter is uninteresting and is called the **trivial solution**, and thus:

$$\begin{aligned} H(0) &= 0 \\ H(L) &= 0 \end{aligned}$$

2.2.2 Time-Dependent Ordinary Differential Equation

We first solve the ODE for $G(t)$:

$$\frac{dG}{dt} = -\lambda k G$$

Since this is a first-order linear homogeneous ODE with constant coefficients, the solution is an exponential of the form:

$$G(t) = ce^{-\lambda kt}$$

2.2.3 Boundary Value (Eigenvalue) Problem

Next, we solve the second-order ODE for $H(x)$:

$$\begin{aligned}\frac{d^2 H}{dx^2} &= -\lambda H \\ H(0) &= 0 \\ H(L) &= 0\end{aligned}$$

Since there are boundary conditions, this is called a **boundary value problem**.

We try three different scenarios for the **eigenvalue** λ :

- $\lambda < 0$

If we let $\lambda = -\beta^2$, our ODE has solutions of the form:

$$H(x) = c_1 \cosh(\beta x) + c_2 \sinh(\beta x)$$

We then apply the boundary conditions:

$$\begin{aligned}H(0) &= c_1 = 0 \\ H(L) &= c_2 \sinh \beta L = 0 \rightarrow c_2 = 0 \quad (\text{if } \beta \text{ is positive})\end{aligned}$$

Thus, this is trivial and there are no negative eigenvalues.

- $\lambda = 0$

If $\lambda = 0$, we can simply integrate twice and find that solutions are of the form:

$$H(x) = c_1 x + c_2$$

We then apply the boundary conditions:

$$\begin{aligned}H(0) &= c_2 = 0 \\ H(L) &= c_1 L = 0 \rightarrow c_1 = 0\end{aligned}$$

Thus, this is also trivial and there is no eigenvalue of zero.

- $\lambda > 0$

If we let $\lambda = \alpha^2$, then the ODE has solutions of the form:

$$H(x) = c_1 \cos(\alpha x) + c_2 \sin(\alpha x)$$

We then apply the boundary conditions:

$$\begin{aligned}H(0) &= c_1 = 0 \\ H(L) &= c_2 \sin \alpha L = 0 \rightarrow \alpha L = n\pi \quad \text{for the non-trivial case}\end{aligned}$$

Thus, the eigenvalues are:

$$\lambda = \left(\frac{n\pi}{L}\right)^2, \quad n = 1, 2, 3, \dots$$

And the corresponding **eigenfunctions** are:

$$H(x) = \sin \frac{n\pi x}{L}$$

2.2.4 Fourier Series

The overall solution of our PDE is:

$$u(x, t) = \sum_{n=1}^m B_n \sin \frac{n\pi x}{L} e^{-\frac{ktn^2\pi^2}{L^2}}$$

Next, we need to find B_n using the initial condition $f(x)$. In some situations this is easy, but for many functions, it isn't possible if m is finite and requires a **Fourier series**:

$$f(x) = \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L}$$

We know that:

$$\int_0^{\frac{L}{2}} \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx = \begin{cases} 0 & m \neq n \\ \frac{L}{2} & m = n \end{cases}$$

Thus, we can multiply both sides by $\sin \frac{m\pi x}{L}$, integrate, and rearrange to get:

$$\begin{aligned} f(x) \sin \frac{m\pi x}{L} &= \sum_{n=1}^{\infty} B_n \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} \\ \rightarrow \int_0^L f(x) \sin \frac{m\pi x}{L} dx &= \int_0^L \sum_{n=0}^{\infty} B_n \sin \frac{n\pi x}{L} \sin \frac{m\pi x}{L} dx \\ \rightarrow \int_0^L f(x) \sin \frac{m\pi x}{L} dx &= \frac{L}{2} B_m \\ \rightarrow B_m &= \frac{2}{L} \int_0^L f(x) \sin \frac{m\pi x}{L} dx \end{aligned}$$

Since functions are vectors, this is essentially taking the inner product of $f(x)$ and $\sin \frac{m\pi x}{L}$:

$$\langle f(x), \sin \frac{m\pi x}{L} \rangle = \int_0^L f(x) \sin \frac{m\pi x}{L} dx$$

Definition 2.4. Two functions $f_1(x)$ and $f_2(x)$ are **orthogonal** over the interval $[a, b]$ if:

$$\langle f_1, f_2 \rangle = \int_a^b f_1 f_2 dx = 0$$

Since functions of the form $\sin(nx)$ are orthogonal if n is different, the eigenfunctions of our solution are an **orthogonal set of vectors**.

3 Numerical Solutions of PDEs

3.1 Linear Advection Equation

Consider the following linear PDE, called the **advection equation**:

$$\frac{\partial}{\partial t} u(x, t) + U_0 \frac{\partial}{\partial x} u(x, t) = 0$$

We note that the total time derivative is:

$$\frac{d}{dt} = \frac{\partial}{\partial t} + \frac{dx}{dt} \frac{\partial}{\partial x}$$

Therefore, if $\frac{dx}{dt} = U_0$, our PDE is just :

$$0 = \frac{\partial}{\partial t} u(x, t) + \frac{dx}{dt} \frac{\partial}{\partial x} u(x, t) = \frac{d}{dt} u$$

This is effectively saying that u is moving at speed U_0 , and within its frame of reference (along paths given by $\frac{d}{dt}x$), it is constant.

We easily determine that $x(t) = x_0 + U_0 t$ and thus $x_0 = x(t) - U_0 t$, meaning we can convert from a point (x, t) to a point $(x_0, 0)$ with the same value of u and thus that we've reduced our solution to:

$$u(x(t), t) = u(x - U_0 t, 0) = u_0(x - U_0 t)$$

3.1.1 Burger's Equation

The following non-linear PDE is known as **Burger's equation**:

$$\frac{\partial}{\partial t} u + u \frac{\partial}{\partial x} u = 0$$

3.1.2 Introduction to Linearization

Since Burger's Equation is non-linear, it's difficult to solve analytically, but we can approximate it using a technique called **linearization**.

We approximate u with a **reference state** and a **deviation**:

$$\underbrace{u(x, t)}_{\text{Full solution}} = \underbrace{U_0}_{\text{Reference state (constant)}} + \underbrace{\epsilon}_{\text{"Small constant"}} * \underbrace{\tilde{u}(x, t)}_{\text{Perturbation}}$$

Substituting this into the PDE and noting that U_0 is constant and a solution to the PDE gives us:

$$\begin{aligned} & \frac{\partial}{\partial t} (U_0 + \epsilon \tilde{u}) + (U_0 + \epsilon \tilde{u}) \frac{\partial}{\partial x} (U_0 + \epsilon \tilde{u}) = 0 \\ \rightarrow & \underbrace{\frac{\partial}{\partial t} U_0 + U_0 \frac{\partial}{\partial x} U_0}_{U_0 \text{ solution to PDE}} + \epsilon \frac{\partial}{\partial t} \tilde{u} + \underbrace{\epsilon \tilde{u} \frac{\partial}{\partial x} U_0}_{U_0 \text{ constant}} + U_0 \epsilon \frac{\partial}{\partial x} \tilde{u} + \epsilon^2 \tilde{u} \frac{\partial}{\partial x} \tilde{u} = 0 \\ \rightarrow & \epsilon \left[\frac{\partial}{\partial t} \tilde{u} + U_0 \frac{\partial}{\partial x} \tilde{u} + \epsilon \tilde{u} \frac{\partial}{\partial x} \tilde{u} \right] = 0 \\ \rightarrow & \frac{\partial}{\partial t} \tilde{u} + U_0 \frac{\partial}{\partial x} \tilde{u} + \underbrace{\epsilon \tilde{u} \frac{\partial}{\partial x} \tilde{u}}_{\text{non-linear}} = 0 \end{aligned}$$

If $|\epsilon| \ll 1$, then we have the advection equation:

$$\frac{\partial}{\partial t} \tilde{u} + U_0 \frac{\partial}{\partial x} \tilde{u} = 0$$

3.2 Numerical Methods

It is often difficult or impossible to analytically solve non-linear PDEs, and thus we use numerical methods to approximate their solutions. However, numerical methods have the following drawbacks:

- A numerical solution only applies to one set of initial conditions, unlike analytical solutions that apply everywhere
- Numerical solutions (typically) have inherent error, so we need to understand it and keep it within a reasonable range
- Sometimes the error isn't large enough to be glaringly obvious, but still breaks the laws of physics, so numerical methods should be paired with analytical theory like checking conservation of energy.